THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the bulb in the tester does not glow with iron nail, it could be because: 1

(a) the nail is coated with paint or rust **(b)** iron is an insulator
(c) nail is a plastic **(d)** cell is stronger

2. A circuit has a cell, a switch, a bulb and a fuse in one loop. If the fuse melts, the bulb: 1

(a) glows brighter **(b)** stays the same **(c)** glows dimmer **(d)** goes off

3. Which statement about a fuse wire is correct? 1

(a) it is made of rubber
(b) it never melts
(c) its melting point is lower than that of the connecting wires
(d) its melting point is higher

4. Which of these should not be done to identify a live wire? 1

(a) switch off main supply **(b)** use a tester
(c) touch it **(d)** use insulated tools

5. A bulb glows when three cells of 1.5 V are connected in series, but not when one cell is reversed. The reason is: 1

(a) the bulb is faulty
(b) the reversed cell reduces the total voltage
(c) the reversed cell adds voltage
(d) wires are copper

6. A circuit tester lights up when connected across a lemon, but not when connected across a dry wooden stick. This suggests: 1

(a) the bulb is wrong **(b)** the lemon is a metal
(c) wood conducts **(d)** the lemon juice conducts

7. Why are high-voltage power lines placed on tall poles and not buried in ordinary ways? 1

(a) to cool **(b)** to avoid contact and shock
(c) to look nice **(d)** to save wire

8. Which of these would stop a tester from lighting? 1

(a) a closed switch **(b)** a clean bulb
(c) a broken connecting wire **(d)** a new cell

9. **Assertion (A):** Electric wires are covered with plastic or rubber. 1
Reason (R): Copper is a good conductor of electricity.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A bulb glows even when there is a gap in the circuit. 1
Reason (R): Current flows only in a closed circuit.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a bulb connected to a battery of three cells glow brighter than with one cell? 2

12. Why are the bodies of plugs and switches made of plastic? 2

13. Why can a 5 A fuse not be replaced with a 15 A fuse in a lighting circuit? 2

14. Why is a switch connected in series with an appliance? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student wants to make a battery of 9 V using 1.5 V cells. How many cells are needed and how are they joined? 3

16. A fuse of 10 A blew while using many appliances. Explain what to do. 3

17. Explain why batteries in a toy are removed when it is not used for a long time. 3

OR

Why does a bulb connected to a battery of cells in the wrong order not glow? 3

18. Explain how LED bulbs help reduce power plants' pollution. 3

OR

Why do we switch off lights and fans when leaving a room? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In Mrs Verma's house in Jaipur, many heavy appliances were switched on together on one socket. The wire became hot and, suddenly, the electricity in that room went off. The electrician found that a fuse had blown. 5
(i) Name the safety device that stopped the current.
(ii) What property of fuse wire makes it suitable?
(iii) Name the effect of current that made the wire hot.
(iv) Why should an ordinary copper wire not be used in place of the fuse wire?

20. Explain electrical safety in a school laboratory including wiring, fuse, switches and first aid. 5

OR

Describe how the invention of electric cell and bulb changed life, and the role of safety rules. 5

END OF QUESTION PAPER — **Total: 40 Marks**